

Animal Biology

Land Sciences

May, 2026



**SE
TU**

Ollscoil
Teicneolaíochta
an Oirdheiscirt

South East
Technological
University

Short Title: Animal Biology
Faculty Science and Computing
Department Land Sciences
Cluster Agriculture
Author VNYHAN
Credits 5

Level: Introductory

Description of Module / Aims

This module is designed to equip students with a introductory understanding of animal physiology, including the structure and function of the animal cell, various animal organs and systems. Students will learn about the primary meat cuts and how to guarantee high quality meat post-slaughter. The student will also gain an insight into the basic properties of feed nutrients and associated nutritional-related disorders, and will examine the foremost animal feeds used in the main agricultural enterprises in Ireland. The module also explores the basics of microbiology. The student will learn to operate safely in a laboratory, performing dissection-based explorations of animal anatomy.

Programmes

BIOL-0019	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 2 / M
BIOL-0019	BSc in Agriculture (WD_SAGUL_D)	1 / 2 / M
BIOL-0019	BSc in Agriculture (WD_SAGUL_D)	2 / 4 / E

Indicative Content

- Animal physiology: structure and function of animal cell and main body organs, e.g. heart, lungs, kidneys, liver, immune system
- Living processes in animals: monogastric and ruminant digestive; reproduction; cellular respiration and enzyme activity
- Meat quality: meat cuts and changes in meat post-slaughter
- Animal nutrition: common foods and classification; food analysis tables; energy and protein requirements; diseases and disorders associated with animal nutrition
- Microbiology: characteristics of microorganisms; role of microorganisms in causing disease and in food production

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the structure and function of the animal cell, cell organelles, the heart, lungs, liver, kidney and relate to animal health.
2. Describe the structure and function of the male and female reproductive tracts, and discuss animal reproduction.
3. Outline the structure and function of the monogastric and ruminant digestive tract, and relate to the management of animal nutrition.
4. Discuss the basic properties, function of feeds and feed requirements in animals, including diseases and disorders associated with animal diet.
5. Describe the slaughtering process and outline the factors that influence the quality of meat post-slaughter.
6. Discuss how enzymes function and explain cellular respiration.
7. Describe microorganisms and discuss microbiology as it relates to agriculture and food production.
8. Operate safely in a laboratory and perform basic laboratory experiments with animal tissues and maintain appropriate records of lab activities.

Learning and Teaching Methods

- Lectures.
- Laboratory dissections of animal organs.
- Laboratory demonstrations of different feed types.
- Video/CD presentations.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,6,7
Continuous Assessment	50%	
<i>Practical</i>	50%	8

Assessment Criteria

- <40%:** Inadequate understanding of the fundamentals of the learning outcomes and insufficient competence in animal biology.
- 40%-49%:** Ability to show a basic knowledge of the key learning outcomes including animal physiology and living processes, as well as be able to perform simple laboratory experiments with animals.
- 50%-59%:** As well as the above the student will be able to demonstrate an understanding of some of the complexities of the topics, and predict nutrition requirements of the animal and potential nutritional disorders. The student will also be able to perform good quality experiments with animal tissues.
- 60%-69%:** In addition, the student will demonstrate more detailed knowledge of the all topics and demonstrate an ability to apply detailed knowledge acquired in the module to new material. The student will also be able to perform very good quality experiments with animal tissues .
- 70%-100%:** The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, predict nutrition requirements of the animal and potential nutritional disorders and show the ability to evaluate and relate topics that overlap all of the learning

outcomes. The student will also be able to perform high quality experiments with animal tissues.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- "Animal Health Ireland." <http://www.animalhealthireland.ie/>
- Blowey , R.W. _A veterinary book for dairy farmers_. Sheffield: 5m publishing, 1999.
- Mc Mangle , et al. _Introduction to farm animals_. Piltown: Teagasc, 2008.
- McDonald, P., R.A. Edwards, J.F.D. Greenhalgh and C.A. Morgan. _Animal nutrition_ . 6 Edition. Essex: Prentice Hall, 2002.
- Roberts, M.B.V. _Biology -A Functional Approach_. 4th ed. Surrey, UK: Nelson, 1986.
- Soffe, R.J (editor). _the agricultural notebook_. 20th ed.. Oxford: Blackwell publishing, 2003.
- Tortora, G.J., B.R. Funke and C.L. Case. _Microbiology -An introduction_. 5th ed. New York: Benjamin/Cummings, 1995.

Requested Resources

- Room Type: Biology Lab